



End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 2421161

Name of the Program and semester: BCA 4th sem

Name of the Course: Computer Organization

Course Code: TBC 403

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

- Q1. (2X10=20 Marks)
- a. List and briefly define the main structural components of a computer. CO1
 - b. What is the difference between computer organization and computer architecture? Discuss their roles with suitable examples. CO1
 - c. Explain history of Computer through different generations. Highlight key technological advancements in each generation. CO1
- Q2. (2X10=20 Marks)
- a. Explain the fetch-decode-execute cycle in detail with example. CO3
 - b. Explain the control unit and microinstructions. Differentiate between hardwired and microprogrammed control units. CO3
 - c. Explain the memory hierarchy and discuss how it affects system performance. CO2
- Q3. (2X10=20 Marks)
- a. Consider a cache with 256 lines of 16 words is provided with 64 KB configurations for a system, how the address is divided or what is the size of the tag in CO4
 - a) Direct mapping
 - b) Associative mapping
 - c) Set associative mapping (2-way mapping)
 - b. Describe cache memory organization and its working principle with different mapping techniques. CO4
 - c. What is the concept of virtual memory? How does it improve system efficiency? CO4
- Q4. (2X10=20 Marks)
- a. Describe various I/O addressing and control methods. CO5
 - b. Explain interrupts and DMA. Discuss their roles in improving system performance. CO5
 - c. Explain different I/O devices and their interfaces. CO5
- Q5. (2X10=20 Marks)
- a. Explain the concept of pipelining in processors with an example. CO5
 - b. Differentiate between RISC and CISC architectures. CO5
 - c. Explain various memory addressing modes. CO5